



ABOUT ME

Civil engineer in spirit, researcher by trade. I make software that plans liver tumor ablations: where needles go, how heat spreads, what to protect. The math is heavy, so I move it to GPUs and then make the interface calm enough for an operating room.

Outside the lab I build games (Sensha Wars in Godot), 3D models for museums, and I run my own GPU inference cluster at home. That last one gets me strange looks at conferences. Worth it.

TOOLKIT

Languages

Python C/C++ CUDA

Medical imaging

3D Slicer PyTorch VTK
OpenIGTLink nnU-Net

Graphics & XR

OpenGL Unity Godot
Blender Maya Meta Quest

Ops

Docker
GPU serving (vLLM, SGLang) Linux

LANGUAGES

German native, Abitur 1.2
French native
English C2, LanguageCert Level 3 (2022)

CERTIFICATES

LanguageCert Test of English C2, April 2022

FAST FACTS

Built a tank RTS as my Baccalaureat project. It had AI opponents and resource management.

Worked two years as caterer and village shop bookkeeper while studying.

Jonas Mehtali

PhD Student, Computer-Assisted Interventions

I make surgical planning so fast and intuitive that the surgeon's intuition becomes the bottleneck.

Email me

Website

GitHub

LinkedIn

ORCID

Scholar

01. EXPERIENCE

PhD Researcher

Sep 2024 - present

ICube (IMAGeS), University of Strasbourg / IHU Strasbourg

- Multi-needle planning for thermal ablation of liver tumors, where manual planning leaves 10-39% local recurrence within five years. I build the optimization that closes that gap: exhaustive search, learning-based estimation, and the decision math between them.
- HEAT, a GPU multi-resolution ablation simulator, plans a full treatment in under a second. C-NCA, a chained neural cellular automaton, estimates cell death with 12,210 parameters at 476 fps.
- Intraoperative replanning: track inserted needles, recompute the plan as the patient breathes, and keep the interface usable by clinicians, validated through studies with medical experts.

End-of-Studies Internship

Jan - Aug 2024

ICube (IMAGeS) / IHU Strasbourg

- Real-time ablation simulation running on networked GPU compute, with co-planning across three concurrent mixed-reality clients (Meta Quest) over a secure VPN.
- Wrote a custom networking protocol with modified-Huffman compression so the OR keeps zero extra hardware.

Research Internship

Jun - Sep 2023

TU Darmstadt (MEC-Lab) / ICube (IMAGeS)

- Took over the CryoTrack cryoablation navigation system: breathing synchronization, EM tracking, and a targeting ring that corrects needle orientation in real time.
- Phantom study in the OR at CHU Strasbourg. Result: accepted paper at MICCAI 2024.

Research Camp

May - Jul 2022

University of Oviedo, Spain

- Full research arc in eight weeks, from hypothesis to published paper: how serif vs sans serif fonts affect e-commerce usability (PeerJ Computer Science, 24 citations).

Research Internship

Jun - Jul 2020

EOST, University of Strasbourg

- Built a landslide monitoring tool (Python, Matplotlib) for the La Valette site; it was taken over by an EOST student after my internship.

02. PUBLICATIONS AND PATENT

- C-NCA: Chained Neural Cellular Automata for Fast and Accurate Thermal Ablation Estimation** **First author** MICCAI 2025, Seoul. 12,210 parameters, 476 fps.
- HEAT: High-Efficiency Simulation for Thermal Ablation Therapy** **First author** IJCARS 2025, 20(6):1135-1143.
- Cryotrack: Planning and Navigation for Computer Assisted Cryoablation** MICCAI 2024, Marrakesh. With H. Krumb, J. Verde, A. Mukhopadhyay, C. Essert.
- How does serif vs sans serif typeface impact the usability of e-commerce websites?** PeerJ Computer Science 2022, with S. Vecino, J. de Andres and the Oviedo HCI group.
- Method and apparatus for determining a thermal ablation volume** **Patent** filed with C. Essert and J. Verde, EP application.

A landslide monitoring tool I wrote is still used by students at EOST Strasbourg.

My thermal simulation estimates 25 min of treatment in 2 ms per frame at 476 fps, on 12,210 parameters.

03. EDUCATION

PhD in Computer-Assisted Interventions

2024 - present

[ICube](#), [University of Strasbourg](#) · [ITI HealthTech](#), [ED269 MSII](#)

MSc Computer Science, Image and 3D

2022 - 2024

[University of Strasbourg](#) · [M1 12.9/20](#), [M2 13.7/20](#)

BSc Computer Science, CMI engineering track

2019 - 2022

[University of Strasbourg](#) · [14.3/20](#), [image processing](#), [computer graphics](#) and [game development curriculum](#)

Institut Sainte-Jeanne d'Arc, Mulhouse

2016 - 2019

[Baccalaureat General 15.4/20](#) and [Allgemeine Hochschulreife \(German Abitur\) 1.2](#)

04. TEACHING AND SERVICE

- Teaching in the Computer Science department at University of Strasbourg: image processing, computer vision, medical imaging, 3D graphics, scientific programming (Python and C++).
- Supervision of PPE1 student projects; mentoring student interns in the IMAGEs group.

05. EARLIER WORK (YES, REALLY)

Caterer and event staff, University of Strasbourg SVU (2022-2024)

General agent and village shop sales and accounting, Town of Niffer (2020-2022)

IT maintenance intern, Concept Office Mulhouse (2018)

Last updated August 2026. Built by hand, as a PDF, by me. All links are clickable, which is exactly the kind of detail that matters.